

# BOUNDARY FIBERS OF FREE DE MORGAN AND KLEENE ALGEBRAS: BOOLEAN INTERVALS, SPERNER EXTREMALITY, AND KALMAN’S THREE GRADES

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ABSTRACT. In the free De Morgan algebra on  $n$  generators, the *boundary* of an element is the tuple of its  $2n$  face restrictions. We determine the fibers of the boundary map completely: each fiber is an interval which is a Boolean lattice  $2^D$ , where  $D$  is an antichain of *diagonal* points of the underlying literal cube; every such antichain arises, so by Sperner’s theorem the largest fiber has exactly  $2^{\binom{n}{\lfloor n/2 \rfloor}}$  elements. Every compatible sphere of faces is a boundary, boundaries biject with antichains of *face* points, and fibers are contracted by an explicit median interpolation in one fresh generator. We then analyze how the least-element retraction interacts with homomorphisms of free algebras: a one-step *defect formula* reaches the fiber minimum, and an exact pointwise *coherence criterion* determines when a substitution respects the least representative of a given fiber. Passing to the subvarieties — by Kalman’s theorem, exactly the Boolean and Kleene varieties — we identify the free Kleene algebra with the monotone functions on the *unmixed* part of the literal cube, recovering the free spectra of Berman–Mukaidono, and show that Kleene boundary fibers have at most two elements, with exactly two non-singleton fibers in every dimension; the Kleene boundary counts coincide with Berman’s constant-free free spectrum (OEIS A007154), giving it a new interpretation. A completeness theorem closes the classification along the signature axis: nontrivial fibers require both the involution and a connection, and once both are present Kalman’s ladder leaves exactly three grades of fiber behaviour — trivial, two polar cells, Sperner-extremal. All finite claims are verified by reproducible exhaustive computations.

## 1. INTRODUCTION

A De Morgan algebra is a bounded distributive lattice equipped with an involution  $\neg$  interchanging  $\wedge, \vee$  and  $0, 1$ ; unlike a Boolean algebra it does not satisfy the law of the excluded middle, so  $x \wedge \neg x$  need not vanish. The finitely generated free De Morgan algebras  $\text{DM}_n$  are classical objects [1, 11, 13, 14]: as a lattice,  $\text{DM}_n$  is the free bounded distributive lattice on the  $2n$  *literals*

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$x_1, \neg x_1, \dots, x_n, \neg x_n$ , hence is isomorphic to the lattice of monotone Boolean functions of  $2n$  variables, of cardinality the Dedekind number  $M(2n)$ . By a classical theorem of Kalman [12], the variety of De Morgan algebras has exactly two proper nontrivial subvarieties: Boolean algebras, and the *Kleene* algebras satisfying  $x \wedge \neg x \leq y \vee \neg y$ .

This paper studies a natural map on these free algebras that appears not to have received attention in the lattice-theoretic literature: the *boundary map*. Substituting an endpoint  $e \in \{0, 1\}$  for a generator  $x_i$  (hence  $1 - e$  for  $\neg x_i$ ) defines a surjective homomorphism  $\partial_i^e : \text{DM}_n \rightarrow \text{DM}_{n-1}$ , the *face maps*; the boundary of  $\varphi$  is the tuple  $\partial\varphi = (\partial_i^e \varphi)_{i \leq n, e \in \{0, 1\}}$  of all its faces. Three questions organize the paper. *Which* boundaries occur, and what is the structure of a *fiber* — the set of all elements with a given boundary? How does the passage to the least element of a fiber interact with homomorphisms of free algebras? And how does the whole theory vary over Kalman’s lattice of subvarieties, and over weaker signatures?

**I. The De Morgan fiber theorem.** Call a point of the literal cube  $\{0, 1\}^{2n}$  *diagonal* if, for every  $i$ , its pair of coordinates for  $(x_i, \neg x_i)$  is  $(0, 0)$  or  $(1, 1)$  — the points invisible to every face.

**Theorem** (Theorem 3.4). Every fiber of the boundary map on  $\text{DM}_n$  is an interval  $[m, M]$  whose difference  $D = M \setminus m$  (in the monotone-function representation) is an antichain of diagonal points, and the fiber is a Boolean lattice, canonically isomorphic to  $2^D$ . Conversely, every antichain of diagonal points arises as the  $D$  of some fiber. Consequently the maximal fiber cardinality in  $\text{DM}_n$  is exactly  $2^{\binom{n}{\lfloor n/2 \rfloor}}$ .

The appearance of Sperner’s theorem [21] is, to our knowledge, new in this context: the diagonal points form a copy of the Boolean cube  $\{0, 1\}^n$ , and the extremal fibers are governed by its maximal antichains. The proof is short and entirely elementary — the crux is that between any two comparable diagonal points there lies a face point — but the statement is sharp, and the machine census shows the Boolean structure is far from sparse: at  $n = 3$  there are 2,130,934 non-singleton fibers, and the explicit complement  $g \mapsto m \vee (M \wedge g^c)$  witnesses Boolean-ness in every one of them. Around the fiber theorem we prove a *filling theorem* (every compatible sphere of prospective faces is a boundary, with a canonical least filler), a *classification of boundaries* (boundaries biject with antichains of *face* points, a Dedekind-type enumeration whose values 4, 82, 4,829,136 appear to be new), and a *median contraction*: for  $\varphi, \varphi'$  in one fiber the element

$$\chi = (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \in \text{DM}_{n+1}$$

in one fresh generator restricts to  $\varphi$  at  $k = 0$ , to  $\varphi'$  at  $k = 1$ , and is constant in  $k$  on every face — a cylinder connecting any two elements of a fiber relative to their common boundary, inside the free algebra. The ternary shape is the Birkhoff–Kiss median [5] adapted to the involutive setting, and exhibits each fiber as a median-closed set [2].

**II. Minimal elements under homomorphisms.** The least element of a fiber is a canonical representative; how canonical is it? For  $g \in \text{DM}_n$  let  $A_g$  be the antichain of minimal *diagonal* points of  $g$ . The *defect formula* (Theorem 6.1) states that one deletion step

$$r(g) = g \setminus A_g$$

already reaches the least element of the fiber of  $g$ . The *coherence dichotomy* (Theorem 6.3) then characterizes exactly when a substitution  $f^* : \text{DM}_n \rightarrow \text{DM}_m$  (the homomorphism induced by an  $n$ -tuple  $f$  of elements of  $\text{DM}_m$ ) is *coherent* with the least-element retraction at a given element:  $r(f^*g) = r(f^*r(g))$  iff the induced map of literal cubes sends no face point into  $A_g$ . The criterion is equational and sharp: substitutions built from faces, permutations and the involution always satisfy it, while degeneracies, connections and diagonal substitutions each violate it on explicitly constructed elements — at  $n = 3$ , on about 14.5 % of all 7,828,354 elements. Finally, the *retracts* of the finitely generated free algebras are determined (Theorem 7.2): they are the finite projective De Morgan algebras, characterized combinatorially by Bova–Cabrer [6] in the unification-theoretic literature — a bridge between two literatures that appear never to have cited one another.

**III. The Kleene interpolation.** Kalman’s theorem bounds what can happen in subvarieties, and the answer is asymmetric in a way we find striking. Call a point of the literal cube *mixed* if some coordinate pair equals  $(0, 0)$  and another equals  $(1, 1)$ , and let  $U_n$  be the subposet of unmixed points. The representation theorem (Theorem 8.2) identifies the free Kleene algebra  $\text{KL}_n$  with the lattice of monotone Boolean functions on  $U_n$  — the Kleene law deletes from the De Morgan representation exactly the mixed points. The cardinalities  $6, 84, 43918, \dots$  recover the free spectra of Berman–Mukaidono [4] (OEIS A268534), an independent confirmation. The fiber theory follows in one stroke (Theorem 9.2): the only unmixed points invisible to the faces are the two poles, they are comparable, so *Kleene boundary fibers have at most two elements, with exactly two non-singleton fibers in every dimension*, and realizable boundaries number exactly  $|\text{KL}_n| - 2$ . Numerically this is  $4, 82, 43916, \dots$  — which is in the OEIS as A007154, *Berman’s constant-free free spectrum of Kleene algebras* [3]; the coincidence is explained by a two-line bijection, giving the classical sequence a new interpretation as a count of compatible spheres. The defect formula and the coherence dichotomy port verbatim, with defects confined to the poles; the summary table:

| theory                | invisible pts   | fibers                         | non-singleton fibers                   |
|-----------------------|-----------------|--------------------------------|----------------------------------------|
| distributive, Boolean | none            | 1                              | none                                   |
| <b>Kleene</b>         | the two poles   | $\leq 2$                       | <b>exactly 2, every <math>n</math></b> |
| De Morgan             | $2^n$ diagonals | $\leq 2^{\lfloor n/2 \rfloor}$ | 65.5 % of elts. of $\text{DM}_3$       |

The Kleene row sits strictly — and very asymmetrically — between its neighbours: the Kleene law does not restore unique fillers, but it cuts the exceptional locus from a positive fraction of all elements down to exactly two fibers, independent of dimension.

**IV. Completeness across signatures.** One may still ask whether *weaker signatures* — interval theories dropping the involution or the connections — realize intermediate fiber behaviours. They do not (Theorem 10.2): over every signature not containing both the involution and a connection, all boundary fibers of the free algebras are singletons. Nontrivial fibers are born only from the interaction of the involution with a connection, and once born they are exhausted by Kalman’s three grades: trivial, two polar cells, Sperner-extremal.

**Motivation, and relation to the literature.** The boundary map arises naturally in the semantics of cubical type theory [8], where the free De Morgan algebra is the interval theory, elements of  $\text{DM}_n$  reparametrize  $n$ -cubes, and the fiber of a boundary describes all *strict* (judgmental) fillers of a fixed cubical boundary; the fiber structure, median interpolation and coherence behaviour were first observed, machine-checked, in a proof-assistant kernel built by the author [19, 20]. The present paper isolates the underlying lattice theory, which is independent of type theory and, we believe, of interest in its own right; the results are stated and proved purely algebraically. In the recent literature, Doré–Cavallo–Mörtberg [10] automate the *search* for a filler of a given boundary over the same algebras; their representation of candidate solutions is deliberately lossy about the solution *set*, and they isolate, in two sentences of prose on “consistent” assignments [10, §4.3], the combinatorial subtlety specific to the De Morgan case. Our results resolve that subtlety into a complete structure theorem: the unconstrained diagonal points are exactly the free Boolean degrees of freedom. Enumerative prior art on interval sizes in the monotone-function lattice is due to Pawelski [17]; our fibers are particular intervals selected by the boundary conditions. The projectivity characterization we use is due to Bova–Cabrer [6]; the free-spectrum counts to Berman and Berman–Mukaidono [3, 4]. All finite combinatorial claims are verified by reproducible exhaustive computations (§11).

## 2. PRELIMINARIES

**2.1. Free De Morgan algebras as monotone functions.** A *De Morgan algebra*  $(A, \wedge, \vee, 0, 1, \neg)$  is a bounded distributive lattice with a unary  $\neg$  satisfying  $\neg\neg a = a$  and  $\neg(a \wedge b) = \neg a \vee \neg b$ . We write  $\text{DM}_n$  for the free De Morgan algebra on generators  $x_1, \dots, x_n$ . The following classical description is the combinatorial engine of everything below; it is stated in [1, Ch. XI] and [11], and in sharpest form in [14, Thm. 3.2, Cor. 2.5] (cf. [13]).

Let  $Q_n := \{0, 1\}^{2n}$  be the *literal cube*, with coordinates grouped in pairs; a point is  $\alpha = (\alpha_1, \dots, \alpha_n)$  with  $\alpha_i \in \{0, 1\}^2$ , the  $i$ -th pair recording prospective values of  $(x_i, \neg x_i)$  *independently*. Order  $Q_n$  componentwise. Then  $\text{DM}_n$  is isomorphic, as a bounded lattice, to the lattice of monotone functions  $Q_n \rightarrow \{0, 1\}$ , equivalently of *up-sets* of  $Q_n$ , with the generators represented by the coordinate projections; the involution is  $(\neg\varphi)(\alpha) = 1 - \varphi(\sigma\bar{\alpha})$ , where  $\bar{\alpha}$  complements all coordinates and  $\sigma$  swaps each pair. In particular  $|\text{DM}_n| = M(2n)$ , the Dedekind number of monotone functions on  $2n$  variables [16]. We identify elements with their up-sets and write  $\varphi \subseteq \psi$  for the lattice order,  $\varphi^c$  for the *pointwise* complement of the up-set (a down-set, in general not an element of  $\text{DM}_n$ ).

**2.2. Faces, boundaries, and the two kinds of points.** For  $i \leq n$  and  $e \in \{0, 1\}$  the *face map*  $\partial_i^e : \text{DM}_n \rightarrow \text{DM}_{n-1}$  substitutes  $x_i := e$ , i.e. fixes the  $i$ -th pair to  $(0, 1)$  if  $e = 0$  and to  $(1, 0)$  if  $e = 1$ , and forgets it. Face maps are surjective lattice homomorphisms and satisfy the evident cubical identities  $\partial_j^\delta \partial_i^\varepsilon = \partial_i^\varepsilon \partial_j^\delta$  (indices adjusted). The *boundary* of  $\varphi$  is  $\partial\varphi := (\partial_i^e \varphi)_{i,e}$ , and the *fiber* of a boundary is the set of all  $\psi$  with  $\partial\psi = \partial\varphi$ .

Call  $\alpha \in Q_n$  a *face point* if some pair  $\alpha_i \in \{(0, 1), (1, 0)\}$ , and a *diagonal point* if every pair  $\alpha_i \in \{(0, 0), (1, 1)\}$ ; write  $\text{Face}(Q_n)$  and  $\text{Diag}(Q_n)$ . Face points are exactly the points lying in some face subcube — the points that some face map sees — while the  $2^n$  diagonal points form a copy of the Boolean cube  $\{0, 1\}^n$  (send  $(1, 1) \mapsto 1$ ,  $(0, 0) \mapsto 0$ ), invisible to every face. This dichotomy, specific to the paired (involutive) situation, is what makes the De Morgan case genuinely different from a free distributive lattice.

More generally, an  $n$ -tuple  $f = (t_1, \dots, t_n)$  of elements of  $\text{DM}_m$  induces the *substitution* homomorphism  $f^* : \text{DM}_n \rightarrow \text{DM}_m$ ,  $x_i \mapsto t_i$  (every homomorphism between the free algebras arises this way), and the *literal-cube map*  $F : Q_m \rightarrow Q_n$ ,  $F(\alpha)_i = (t_i(\alpha), (\neg t_i)(\alpha))$ , so that  $(f^*g)(\alpha) = g(F(\alpha))$ . Note that  $F$  is a map of *all* points and need not preserve the face/diagonal dichotomy; this freedom is where the coherence analysis of §6 happens. Composing tuples by substitution makes the finitely generated free algebras into a category; we occasionally use the language of the opposite category, the *De Morgan cube category*, whose objects are the formal cubes  $[n]$  and whose morphisms  $[m] \rightarrow [n]$  are the tuples  $f$  above.

**Lemma 2.1.** *If  $\alpha < \beta$  are comparable diagonal points, there is a face point  $\gamma$  with  $\alpha < \gamma \leq \beta$ .*

*Proof.* Some pair has  $\alpha_i = (0, 0)$  and  $\beta_i = (1, 1)$ . Let  $\gamma$  agree with  $\alpha$  except  $\gamma_i := (0, 1)$ . Then  $\alpha < \gamma$ ,  $\gamma \leq \beta$ , and  $\gamma$  is a face point.  $\square$

### 3. THE STRUCTURE THEOREM

**Lemma 3.1.** *Every fiber is an interval: it is closed under  $\wedge, \vee$  (hence has a least element  $m$  and greatest element  $M$ ), and every  $\psi$  with  $m \subseteq \psi \subseteq M$  lies in the fiber.*

*Proof.* Faces are lattice maps, so fibers are sublattices; finiteness gives  $m, M$ . If  $m \subseteq \psi \subseteq M$  then  $\partial_i^e m \subseteq \partial_i^e \psi \subseteq \partial_i^e M$  by monotonicity of  $\partial_i^e$ , and the outer terms are equal.  $\square$

**Lemma 3.2.** *For a fiber  $[m, M]$ , the difference  $D := M \setminus m$  contains no face point.*

*Proof.* A face point of  $M \setminus m$  would make the corresponding faces of  $M$  and  $m$  differ.  $\square$

**Lemma 3.3.**  *$D$  is an antichain.*

*Proof.* Suppose  $\alpha < \beta$  in  $D$ ; both are diagonal by Lemma 3.2. Pick  $\gamma$  as in Lemma 2.1. Since  $\alpha \in M$  and  $M$  is an up-set,  $\gamma \in M$ ; since  $\gamma$  is a face point,  $\gamma \notin D$ , so  $\gamma \in m$ ; since  $m$  is an up-set and  $\gamma \leq \beta$ ,  $\beta \in m$ , contradicting  $\beta \in D$ .  $\square$

**Theorem 3.4** (Structure theorem). *Let  $F = [m, M]$  be a fiber of the boundary map on  $\text{DM}_n$  and  $D = M \setminus m$ . Then  $D$  is an antichain of diagonal points, and*

$$2^D \longrightarrow F, \quad S \mapsto m \cup S$$

*is a lattice isomorphism; in particular  $F$  is a Boolean lattice, with complementation  $g \mapsto m \vee (M \wedge g^c)$ . Conversely, for every antichain  $D_0$  of diagonal points there is a fiber with  $D = D_0$  (for  $n \geq 1$ ). Consequently*

$$\max_{\text{fibers of DM}_n} |F| = 2^{\lfloor n/2 \rfloor}.$$

*Proof.*  $D$  is a diagonal antichain by Lemmas 3.2–3.3. For  $S \subseteq D$ , the set  $m \cup S$  is an up-set: a point above some  $\alpha \in S$  is either  $\alpha$  itself or lies above a face point  $\gamma > \alpha$  (Lemma 2.1 if diagonal; itself if a face point, since  $\alpha$ 's strict upper covers in  $Q_n$  raise one coordinate, breaking a  $(0, 0)$  pair to  $(0, 1)$  or  $(1, 0)$ ), and such  $\gamma \in M$ , being a face point, lies in  $m$ ; up-closure of  $m$  finishes. Since  $S$  avoids all face subcubes,  $\partial(m \cup S) = \partial m$ , so  $m \cup S \in F$ ; conversely each  $g \in F$  equals  $m \cup (g \setminus m)$  with  $g \setminus m \subseteq D$ . The correspondence preserves unions and intersections, hence is a lattice isomorphism onto  $F$ ; Boolean complementation transports to the stated formula.

*Realizability.* For  $D_0 = \emptyset$  take the fiber of the generator  $x_1$  (here  $n \geq 1$  is used): all its minimal points are facial, and no diagonal point can be adjoined — adjoining  $\perp$  forces the whole cube by up-closure, and adjoining any other diagonal point outside  $x_1$  brings in a face point with first pair  $(0, 1)$ , changing the boundary — so the fiber is a singleton and  $D = \emptyset$ . For  $D_0 \neq \emptyset$ , let  $m_0$  be the up-closure of the set of face points lying strictly above some point of  $D_0$ , and  $M_0 := m_0 \cup D_0$ . As above  $M_0$  is an up-set, no point of  $D_0$  lies in  $m_0$  (a face point above  $\alpha \in D_0$  is strictly above it, and points of  $D_0$  are pairwise incomparable, so up-closure cannot reach  $\alpha$ ), and  $\partial m_0 = \partial M_0$  since  $D_0$  avoids the faces. Moreover  $m_0$  is the least element of its fiber, directly: every element with the same boundary as  $m_0$  contains

the same face points, and  $m_0$  is generated as an up-set by its face points, so every such element contains  $m_0$ .

*Exactness.* The difference  $D$  of this fiber is exactly  $D_0$ . The idea: an additional diagonal point outside  $D_0$  cannot be freely toggled, because its covers being forced into  $m_0$  would put some point of  $D_0$  below a one-coordinate cover of an incomparable diagonal point — impossible. In detail:  $D \supseteq D_0$ , and  $D$  is an antichain containing  $D_0$ , so a hypothetical further point  $\delta \in D \setminus D_0$  is a diagonal point incomparable to every element of  $D_0$ . Every cover of  $\delta$  in the literal cube raises one coordinate of a  $(0,0)$ -pair, hence is a face point; as  $\delta$  lies in the fiber's greatest element, whose face points all lie in  $m_0$ , each such cover  $\gamma = \delta + e_c$  lies in  $m_0$ , so  $\gamma \geq \gamma' > \alpha$  for some face point  $\gamma'$  and some  $\alpha \in D_0$ , whence  $\alpha \leq \gamma$ . Since  $\alpha$  is incomparable with  $\delta$  but  $\alpha \leq \gamma = \delta + e_c$ , the point  $\alpha$  must exceed  $\delta$  precisely at the raised coordinate  $c$ , so  $\alpha_c = 1$ ; being diagonal,  $\alpha$  then has  $(1,1)$  at the raised pair — but  $\gamma$  has  $(0,1)$  or  $(1,0)$  there, contradicting  $\alpha \leq \gamma$ . So no such  $\delta$  exists and  $D = D_0$ .

The maximum: diagonal points form a copy of  $\{0,1\}^n$ , whose maximal antichains have  $\binom{n}{\lfloor n/2 \rfloor}$  elements by Sperner's theorem [21]; both bound and attainment follow.  $\square$

**Remark 3.5.** The proof of realizability is constructive and cheap to check by machine; at  $n = 4$ , the middle layer of the diagonal cube is an antichain of size  $\binom{4}{2} = 6$  and the construction produces a fiber isomorphic to  $2^6$  with all 64 members sharing their boundary — see §11.

**Corollary 3.6.** *All fiber cardinalities in  $\text{DM}_n$  are powers of two, at most  $2^{\binom{n}{\lfloor n/2 \rfloor}}$ ; and  $M(2n) = \sum_{\text{fibers}} 2^{|D|}$ .*

#### 4. EVERY COMPATIBLE SPHERE FILLS

**Definition 4.1.** A *sphere* in  $\text{DM}_n$  is a family  $g = (g_i^e)_{i \leq n, e \in \{0,1\}}$ ,  $g_i^e \in \text{DM}_{n-1}$ , satisfying the compatibility conditions  $\partial_j^\delta g_i^\varepsilon = \partial_i^\varepsilon g_j^\delta$  (for  $i \neq j$ , indices adjusted as usual).

**Theorem 4.2.** *Every sphere is the boundary of an element of  $\text{DM}_n$ ; a canonical filler is the up-closure  $m_g$  of the union of the prescribed 1-points, which is also the least filler.*

*Proof.* Identify each  $g_i^e$  with its up-set inside the face subcube  $Q_n^{(i,e)} \subseteq Q_n$ , and let  $m_g := \uparrow(\bigcup_{i,e} g_i^e)$ . We must show  $\partial_i^e m_g = g_i^e$ . The inclusion  $\supseteq$  is immediate. For  $\subseteq$ , let  $\gamma \in Q_n^{(i,e)}$  lie in  $m_g$ , say  $\gamma \geq \delta$  with  $\delta \in g_j^{e'}$ . If  $(j, e') = (i, e)$ , monotonicity of  $g_i^e$  gives  $\gamma \in g_i^e$ . The case  $j = i$ ,  $e' \neq e$  cannot occur: the two prescribed literal pairs  $(0,1)$  and  $(1,0)$  are incomparable, so  $\delta \leq \gamma$  fails. Hence  $j \neq i$ ; let  $\delta'$  agree with  $\delta$  except that its  $i$ -th pair is set to the value prescribed by the face  $(i, e)$ ; from  $\delta \leq \gamma$  and  $\gamma \in Q_n^{(i,e)}$  one checks  $\delta \leq \delta' \leq \gamma$  componentwise, and  $\delta'$  lies in both face subcubes  $(i, e)$  and  $(j, e')$ .

Monotonicity gives  $\delta' \in g_j^e$ , compatibility transports this to  $\delta' \in g_i^e$  (both sides restrict to the same element of  $\text{DM}_{n-2}$  on the common subcube), and monotonicity again gives  $\gamma \in g_i^e$ . Minimality is clear: any filler's up-set must contain every prescribed 1-point.  $\square$

**Theorem 4.3** (Classification of boundaries). *The assignments  $g \mapsto m_g$  and  $\varphi \mapsto \partial\varphi$  are mutually inverse bijections between spheres and the set of elements of  $\text{DM}_n$  all of whose minimal points are face points. Consequently spheres (equivalently, realizable boundaries) are in bijection with antichains of face points of  $Q_n$ , and*

$$B(n) = \#\{\text{antichains of the poset of face points of } Q_n\},$$

*a Dedekind-type quantity for the  $(4^n - 2^n)$ -point poset of face points.*

*Proof.* A least fiber element  $m$  contains no removable point, i.e. no minimal point that is diagonal (removing a minimal diagonal point preserves up-closure and all faces); conversely if every minimal point of  $\varphi$  is a face point then no proper boundary-preserving shrinking exists, and Theorem 4.2 shows every sphere arises. An up-set is determined by its antichain of minimal points; those with all minimal points facial correspond exactly to antichains inside the subposet of face points.  $\square$

**Remark 4.4.** For  $n = 1$  the face points are the two incomparable points  $(0, 1), (1, 0)$ , giving  $B(1) = 4$ ; for  $n = 2$  the twelve face points give  $B(2) = 82$ ; the machine count of antichains at  $n = 3$  (a 56-point poset) confirms  $B(3) = 4,829,136$ , agreeing with the direct census (§11).

## 5. MEDIAN CONTRACTION OF A FIBER

The Boolean structure of Theorem 3.4 is a statement *about* the fiber; one also wants terms *inside the free algebra* connecting two elements of a fiber. One fresh generator suffices.

**Theorem 5.1.** *Let  $\varphi, \varphi' \in \text{DM}_n$  have equal boundaries, and let  $k$  be a fresh generator. Then*

$$\chi := (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \in \text{DM}_{n+1}$$

*satisfies*

- (1)  $\partial_k^0 \chi = \varphi$  and  $\partial_k^1 \chi = \varphi'$ ;
- (2) *for every face  $(i, e)$  of the original generators,  $\partial_i^e \chi$  is the image of the common face  $\partial_i^e \varphi = \partial_i^e \varphi'$  under the inclusion  $\text{DM}_{n-1} \hookrightarrow \text{DM}_n$  adding  $k$  — i.e.  $\chi$  is constant in  $k$  on the entire boundary.*

*Proof.* (1) Substituting  $k := 0$  gives  $\varphi \vee (\varphi \wedge \varphi') = \varphi$ ;  $k := 1$  gives  $\varphi' \vee (\varphi \wedge \varphi') = \varphi'$  (absorption). (2) Writing  $\rho := \partial_i^e \varphi = \partial_i^e \varphi'$ , the face of  $\chi$  is  $(\rho \wedge \neg k) \vee (k \wedge \rho) \vee \rho = \rho$ , again by absorption.  $\square$



**Remark 5.2.** The shape of  $\chi$  is a directed variant of the Birkhoff–Kiss median  $(a \wedge b) \vee (b \wedge c) \vee (a \wedge c)$  [5]; the third term is essential — the naive interpolation  $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$  fails (2) precisely because the free De Morgan algebra does not satisfy the excluded middle for  $k$ . It also exhibits each fiber as a median-closed set: for  $\varphi, \varphi', \psi$  in a fiber, the median of the three lies in the fiber (immediate from Lemma 3.1, and an instance of majority-closure phenomena classical in the theory of median algebras [2]).

## 6. MINIMAL ELEMENTS UNDER HOMOMORPHISMS

The least element of a fiber is a canonical representative for a *fixed*  $n$ ; this section analyzes its interaction with the substitution homomorphisms  $f^* : \text{DM}_n \rightarrow \text{DM}_m$ .

For  $g \in \text{DM}_n$  let  $A_g$  be the set of minimal points of  $g$  (qua up-set) that are diagonal. Minimal points form an antichain, so  $A_g$  is an antichain of diagonal points.

**Theorem 6.1** (defect formula). *For every  $g \in \text{DM}_n$ , the element  $r(g) := g \setminus A_g$  is the least element of the boundary fiber of  $g$ . In particular one deletion step reaches the fiber minimum, and  $g$  is the least element of its fiber iff  $A_g = \emptyset$ .*

*Proof.*  $r(g)$  is an up-set: a point strictly above  $\alpha \in A_g$  is either a face point (hence in  $g$ , hence in  $r(g)$ ), or a diagonal point  $\beta > \alpha$ , in which case Lemma 2.1 provides a face point  $\gamma$  with  $\alpha < \gamma \leq \beta$ ; then  $\gamma \in g$ , so  $\beta \in \uparrow\gamma \subseteq r(g)$ . Its boundary equals  $\partial g$  since only diagonal points were removed. No new diagonal minimal points appear: if  $\beta$  is diagonal and minimal in  $r(g)$ , then  $\beta$  was not minimal in  $g$  (else  $\beta \in A_g$ ), so some  $\delta < \beta$  lies in  $g$ ; taking  $\delta$  minimal in  $g$ , either  $\delta$  is a face point — but then  $\delta \in r(g)$ , contradicting minimality of  $\beta$  — or  $\delta \in A_g$  is diagonal, and Lemma 2.1 gives a face point  $\gamma \in (\delta, \beta]$ , with  $\gamma \neq \beta$  ( $\beta$  is diagonal), lying in  $r(g)$  — again a contradiction. So all minimal points of  $r(g)$  are face points, which characterizes the least element of the fiber (Theorem 4.3).  $\square$

**Corollary 6.2** (the defect of a substitution). *Let  $f^* : \text{DM}_n \rightarrow \text{DM}_m$  be a substitution and  $\varphi \in \text{DM}_n$  the least element of its fiber. Then the deviation of  $f^*\varphi$  from fiber-minimality is the explicit antichain*

$$A_{f^*\varphi} = \{ \alpha \in \text{Diag}(Q_m) : F(\alpha) \in \varphi \text{ and } F(\beta) \notin \varphi \text{ for every lower cover } \beta \triangleleft \alpha \},$$

and  $r(f^*\varphi)$  differs from  $f^*\varphi$  exactly by this antichain, inside one Boolean fiber; the median cylinder of Theorem 5.1 connects the two.

Call  $f$  and  $g \in \text{DM}_n$  *coherent* if  $r(f^*g) = r(f^*r(g))$ .

**Theorem 6.3** (coherence dichotomy). *For every substitution  $f$  and every  $g \in \text{DM}_n$ , the following are equivalent:*

- (1)  $(f, g)$  is coherent;

- (2)  $\partial(f^*g) = \partial(f^*r(g))$ ;
- (3)  $F(\text{Face}(Q_m)) \cap A_g = \emptyset$ .

*Proof.* (3) $\Rightarrow$ (2):  $g$  and  $r(g)$  agree outside  $A_g$ , and the boundaries of the transports only read the values of  $g$ , resp.  $r(g)$ , on  $F(\text{Face}(Q_m))$ . (2) $\Rightarrow$ (1): elements with equal boundaries lie in one fiber, whose least element is unique. (1) $\Rightarrow$ (3), contrapositively: if  $\alpha$  is a face point with  $F(\alpha) \in A_g$ , then  $(f^*g)(\alpha) = g(F(\alpha)) = 1$  while  $(f^*r(g))(\alpha) = 0$ ; a disagreement at a face point is a disagreement of boundaries, and least elements determine their boundaries ( $r(a) = r(b)$  forces  $\partial a = \partial r(a) = \partial r(b) = \partial b$ ), so the two corrected elements differ.  $\square$

**Proposition 6.4** (coherent generators and obstructions). *Faces, permutations of the generators, the involution substitutions  $x_i := \neg x_i$ , and the substitutions  $x_i := x_i \wedge \neg x_i$ ,  $x_i := x_i \vee \neg x_i$  send face points to face points. The class of face-point-preserving substitutions is closed under composition and contains the subcategory generated by these maps; every substitution  $f$  in it is coherent for every  $g$ , so that*

$$r \circ f^* = r \circ f^* \circ r :$$

*the corrected transport  $g \mapsto r(f^*g)$  depends only on the least representative of the source fiber, and it composes functorially on this class:  $r((f'f)^*g) = r(f'^*r(f^*g))$ ; restricted to the sets of fiber-minimal representatives (on which the identity substitution acts as the identity), it thus defines a functor. (Note that  $r$  is not natural in the plain sense  $r \circ f^* = f^* \circ r$ : for  $f^*(x) = x \wedge \neg x$  and  $g = x$ , one has  $r(f^*x) = 0 \neq x \wedge \neg x = f^*(rx)$ .) Degeneracies ( $x_{n+1}$  unused), connections ( $x_i := x_i \wedge x_j$ ) and diagonal substitutions ( $x_i := x_j$ ) each map some face point to a diagonal point, and for each of them elements  $g$  violating condition (3) exist.*

*Proof.* For a face,  $F$  embeds  $Q_{n-1}$  into a face subcube, sending every point (facial or not) to a face point. Permutations and involutions permute pair-coordinates and swap pair entries, preserving the dichotomy. For  $x_i := x_i \wedge \neg x_i$ : if  $\alpha$  is facial via pair  $j \neq i$ , so is  $F(\alpha)$ ; if via pair  $i$ , then  $(x_i \wedge \neg x_i, x_i \vee \neg x_i)$  evaluates at  $(0, 1)$  or  $(1, 0)$  to  $(0, 1)$ , a facial pair. Closure under composition is immediate ( $F'F(\text{Face}) \subseteq F'(\text{Face}) \subseteq \text{Face}$ ). Coherence for every  $g$  is Theorem 6.3(3 $\Rightarrow$ 1); the displayed identity is its restatement, and functoriality follows by applying coherence of  $(f', f^*g)$ :  $r(f'^*r(f^*g)) = r(f'^*f^*g) = r((f'f)^*g)$ . For a degeneracy, a point facial only via the dropped pair maps to a diagonal point; for the connection  $x_1 := x_1 \wedge x_2$ , the point with pairs  $((1, 0), (0, 0), \dots)$  maps to  $((0, 0), (0, 0), \dots)$ ; for the diagonal  $x_1 := x_2$ , the point  $((0, 1), (0, 0), \dots)$  maps to  $((0, 0), (0, 0), \dots)$ . Violating elements exist in each case: take  $g$  with  $A_g$  containing the reached diagonal point (realizability in Theorem 3.4).  $\square$

**Remark 6.5.** The degeneracy case deserves emphasis. For the cylinder  $\sigma^*g$  along the degeneracy  $\sigma^* : \text{DM}_n \rightarrow \text{DM}_{n+1}$ , the two new faces are  $g$  itself;

if  $A_g \neq \emptyset$  then condition (3) fails (every diagonal point of  $Q_n$  is the image of a point facial only via the added pair), and indeed  $r(\sigma^*g) \neq \sigma^*(rg)$ : the minimal element of the cylinder's fiber is not the cylinder of the minimal element. At  $n = 3$  the incoherent locus of a diagonal substitution comprises 1,133,756 of the 7,828,354 elements — about 14.5 % — so the dichotomy is a genuinely quantitative constraint (§11).

## 7. RETRACTS AND PROJECTIVITY

Which algebras arise as retracts of the finitely generated free De Morgan algebras? In any variety, retracts of free algebras are exactly the projectives, and idempotents split in the category of algebras; the following records the answer and its combinatorial content, together with the categorical repackaging that motivated it.

**Proposition 7.1.** *The idempotent endomorphism  $e^*$  of  $\text{DM}_1$  with  $e^*(x) = x \vee \neg x$  does not split through any free algebra: there are no substitutions  $p^* : \text{DM}_k \rightarrow \text{DM}_1$  and  $s^* : \text{DM}_1 \rightarrow \text{DM}_k$  with  $s^*p^* = \text{id}_{\text{DM}_k}$  and  $p^*s^* = e^*$ . Equivalently, the De Morgan cube category is not idempotent complete.*

*Proof.* Idempotency is the absorption  $(x \vee \neg x) \vee \neg(x \vee \neg x) = (x \vee \neg x) \vee (\neg x \wedge x) = x \vee \neg x$ . If a splitting existed,  $\text{DM}_k$  would be a retract of  $\text{DM}_1$ ; since  $|\text{DM}_1| = 6$  and  $|\text{DM}_k| = M(2k) \geq 168$  for  $k \geq 2$ , we must have  $k \leq 1$ . For  $k = 0$ ,  $e^*$  factors through  $\text{DM}_0 = \{0, 1\}$ , so is constant on generators; but  $x \vee \neg x \notin \{0, 1\}$  in  $\text{DM}_1$ . For  $k = 1$ , the retraction forces  $s^*$  to be bijective (finite cardinality), so  $e^* = p^*s^*$  is injective; but  $e^*(x) = x \vee \neg x = e^*(x \vee \neg x)$  while  $x \neq x \vee \neg x$ .  $\square$

**Theorem 7.2.** *The idempotent completion of the category of finitely generated free De Morgan algebras is equivalent to the category of finite projective De Morgan algebras. Its objects are characterized combinatorially by Bova–Cabrer [6, Thm. 11]: under Cornish–Fowler duality [9], a finite De Morgan algebra is projective iff its dual involutive poset  $(P, \leq, i)$  is a nonempty lattice, every  $x \leq i(x)$  admits  $y \geq x$  with  $y = i(y)$ , and the subposet  $\{x \mid x \leq i(x)\}$  is 3-complete. The same holds over the Kleene variety, with the characterization of [6, Thm. 12].*

*Proof.* In any variety, idempotents split in the category of algebras, and retracts of free algebras are exactly the projectives; hence the idempotent completion of the full subcategory of finitely generated free algebras is the full subcategory of finitely generated projectives; conversely, every finite projective De Morgan algebra is finitely generated and is therefore a retract of a finitely generated free algebra, so the comparison is essentially surjective. The varieties of De Morgan and Kleene algebras are locally finite, so finitely generated projectives are finite [6]. The characterizations of the objects are Bova–Cabrer's theorems.  $\square$

**Remark 7.3.** In the language of cubical sets, Proposition 7.1 says the De Morgan cube category is not a generalized Reedy category, in particular

not an Eilenberg–Zilber category, by the same first obstruction Campion [7] exhibited for diagonal–connection sites; Campion’s classification, however, never meets the De Morgan site, since his cube categories are categories of  $\{0, 1\}$ -valued operations, so his full-signature site is the *Boolean* cube category, where  $x \vee \neg x = 1$ . Theorem 7.2 places the De Morgan site in the table of cubical Karoubi envelopes — Dedekind cubes with diagonals complete to finite lattices [18, 7], Boolean cubes to finite nonempty sets [7], De Morgan and Kleene cubes to the duals of finite projective De Morgan, respectively Kleene, algebras — and bridges the cubical literature with the unification-theoretic literature where the projectivity characterization lives [6]; the two appear never to have cited one another.

## 8. THE FREE KLEENE ALGEBRA ON UNMIXED POINTS

A *Kleene algebra* is a De Morgan algebra satisfying  $x \wedge \neg x \leq y \vee \neg y$ ; the variety is generated by the three-element chain  $\mathbf{3} = \{0, u, 1\}$  with  $\neg u = u$  [12, 11]. Write  $\text{KL}_n$  for the free Kleene algebra on  $x_1, \dots, x_n$ . By Kalman’s theorem [12], Boolean and Kleene algebras are the only proper nontrivial subvarieties of De Morgan algebras, so the results of this and the next section, together with §§3–6 and the folklore Boolean case, classify the fiber theory over the whole subvariety lattice.

**Definition 8.1.** A point  $\alpha \in Q_n$  is *mixed* if some pair  $\alpha_i = (0, 0)$  and some pair  $\alpha_j = (1, 1)$ ; otherwise *unmixed*. Let  $U_n \subseteq Q_n$  be the subposet of unmixed points. ( $|U_1| = 4$ ,  $|U_2| = 14$ ,  $|U_3| = 46$ .)

Note that  $U_n$  is closed under the De Morgan symmetry (complement all coordinates, swap each pair), and that the diagonal points of  $Q_n$  that are unmixed are exactly the two poles  $\perp = (0, \dots, 0)$  and  $\top = (1, \dots, 1)$ .

**Theorem 8.2.** *Restriction of monotone functions along  $U_n \hookrightarrow Q_n$  induces an isomorphism from the free Kleene algebra onto the lattice of monotone functions on  $U_n$ , with involution  $(\neg\varphi)(\alpha) = 1 - \varphi(\sigma\bar{\alpha})$  as before:*

$$\text{KL}_n \cong \{\text{monotone } U_n \rightarrow \{0, 1\}\}.$$

*Proof.* The restriction map  $\rho : \text{DM}_n \rightarrow \text{Mon}(U_n)$  is a homomorphism of De Morgan algebras: meets, joins and bounds are pointwise, and  $U_n$  is closed under the symmetry defining  $\neg$ . It is surjective, since a monotone function on  $U_n$  extends to  $Q_n$  (e.g. by its least monotone extension).

$\text{Mon}(U_n)$  is a Kleene algebra. Write  $\theta(\alpha) := \sigma\bar{\alpha}$  for the point involution underlying  $\neg$ ;  $\theta$  fixes the Boolean pairs and exchanges  $(0, 0) \leftrightarrow (1, 1)$ , so for an unmixed  $\alpha$  — whose non-Boolean pairs are all  $(0, 0)$  or all  $(1, 1)$  — the points  $\alpha$  and  $\theta\alpha$  are *comparable*. Now let  $\varphi, \psi$  be monotone on  $U_n$  and suppose  $(\varphi \wedge \neg\varphi)(\alpha) = 1$  and  $(\psi \vee \neg\psi)(\alpha) = 0$  for some  $\alpha \in U_n$ . The first gives  $\varphi(\alpha) = 1$  and  $\varphi(\theta\alpha) = 0$ , so  $\alpha \not\leq \theta\alpha$  and hence  $\theta\alpha < \alpha$  by comparability; the second gives  $\psi(\alpha) = 0$  and  $\psi(\theta\alpha) = 1$ , contradicting monotonicity along  $\theta\alpha < \alpha$ . So  $\varphi \wedge \neg\varphi \leq \psi \vee \neg\psi$  pointwise, for *all*  $\varphi, \psi$ . Thus  $\rho$  factors through the Kleene quotient, giving a surjection  $\text{KL}_n \twoheadrightarrow \text{Mon}(U_n)$ .

For injectivity we use the three-valued semantics. By [11],  $\text{KL}_n$  embeds into  $\mathbf{3}^{(3^n)}$  as the subalgebra generated by the projections; write  $\hat{\varphi} : \mathbf{3}^n \rightarrow \mathbf{3}$  for the function induced by  $\varphi$ . For  $v \in \mathbf{3}^n$  define  $v_{\text{low}}, v_{\text{high}} \in Q_n$  by encoding  $v_i = 0 \mapsto (0, 1)$ ,  $v_i = 1 \mapsto (1, 0)$ , and  $v_i = u \mapsto (0, 0)$  (low), resp.  $(1, 1)$  (high). Both are *unmixed*, because the choice for  $u$  is made uniformly; and a direct induction on terms shows  $\hat{\varphi}(v) = 0, u, 1$  according as the pair  $(\varphi(v_{\text{low}}), \varphi(v_{\text{high}}))$  is  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 1)$  (monotonicity excludes  $(1, 0)$ ): on generators this is the definition of the encoding;  $\wedge$  and  $\vee$  act componentwise on the pairs, whose three achievable values are linearly ordered, matching the chain  $\mathbf{3}$ ; and  $\neg$  sends the pair  $(a, b)$  to  $(1 - b, 1 - a)$  — the point involution  $\theta$  introduced above exchanges  $v_{\text{low}}$  and  $v_{\text{high}}$  — matching negation on  $\mathbf{3}$ . Hence  $\hat{\varphi}$  is determined by  $\rho(\varphi)$ . Conversely every unmixed point is a  $v_{\text{low}}$  or a  $v_{\text{high}}$  (read  $u$  at its diagonal pairs), so  $\rho(\varphi)$  is determined by  $\hat{\varphi}$ ; as  $\varphi \mapsto \hat{\varphi}$  is injective on  $\text{KL}_n$ , so is the induced map  $\text{KL}_n \rightarrow \text{Mon}(U_n)$ .  $\square$

**Remark 8.3.** The counts  $|\text{Mon}(U_n)| = 6, 84, 43918$  for  $n = 1, 2, 3$  (machine enumeration, §11) agree with the free spectra of Kleene algebras computed by Berman–Mukaidono [4] (OEIS A268534), an independent confirmation of Theorem 8.2. The theorem refines the classical normal-form descriptions [11, 4] in the same way the literal-cube picture refines them for De Morgan algebras: it makes the boundary combinatorics visible. In this picture the Kleene law acts by *deleting the mixed points* — and, as the next section shows, mixed points are almost all of the diagonal.

## 9. KLEENE FIBERS: AT MOST TWO ELEMENTS

Faces  $\partial_i^c : \text{KL}_n \rightarrow \text{KL}_{n-1}$  restrict to the face subcubes of  $U_n$  exactly as in the De Morgan case (fixing a pair to  $(0, 1)$  or  $(1, 0)$  cannot create or destroy mixedness). A point of  $U_n$  is a *face point* if some pair is  $(0, 1)$  or  $(1, 0)$ ; the points invisible to every face are the unmixed diagonal points:

**Lemma 9.1.** *The unmixed diagonal points of  $Q_n$  are exactly the poles  $\perp$  and  $\top$ ; they are comparable, and there is a face point strictly between them (any point with one pair  $(0, 1)$  and the rest  $(0, 0)$ ).*

**Theorem 9.2.** *Every boundary fiber of  $\text{KL}_n$  is an interval  $[m, M]$  with  $D = M \setminus m$  an antichain contained in  $\{\perp, \top\}$ ; hence  $|D| \leq 1$  and fibers have at most two elements. Exactly two fibers are non-singleton in every dimension  $n \geq 1$ : the zero-boundary fiber  $\{0, \delta_\top\}$  ( $\delta_\top$  the indicator of  $\top$ , i.e.  $x_1 \wedge \neg x_1 \wedge \cdots \wedge x_n \wedge \neg x_n$ ) and the top-boundary fiber  $\{1 \setminus \perp, 1\}$ . Consequently*

$$B_{\text{KL}}(n) = |\text{KL}_n| - 2,$$

where  $B_{\text{KL}}(n)$  is the number of realizable boundaries.

*Proof.* The interval property, the confinement of  $D$  to invisible points, and the antichain property are the proofs of §3 verbatim, applied to the poset  $U_n$ : the only inputs are that faces see exactly the face points and that between

comparable invisible points there is a face point (Lemma 9.1). Since  $\perp < \top$ , an antichain contains at most one pole. Realizability of  $D = \{\top\}$  and  $D = \{\perp\}$  is witnessed by the displayed fibers (that  $\delta_\top$  is monotone on  $U_n$  and has zero boundary, and dually, is immediate); uniqueness of each — i.e. that no other fiber has  $D \neq \emptyset$  — is direct: if  $D = \{\top\}$  then  $\top \notin m$ , and every nonempty up-set of  $U_n$  contains the global maximum  $\top$ , so  $m = 0$ ; dually, if  $D = \{\perp\}$  then  $\perp \in M$ , and an up-set containing the global minimum is everything, so  $M = 1$  and  $m = 1 \setminus \perp$ . Finally, boundaries biject with fiber-minimal elements (Theorem 4.3), and all elements are fiber-minimal except the top members  $\delta_\top$  and  $1$  of the two polar fibers.  $\square$

**Remark 9.3** (the free spectrum, interpreted). The sequence  $B_{\text{KL}}(n) = 4, 82, 43916, \dots$  appears in the OEIS as A007154, *the free spectrum of Kleene algebras in the constant-free signature* (Berman [3]): dropping the constants  $0, 1$  from the signature removes exactly two elements from each free algebra. Theorem 9.2 gives the same numbers a different meaning — the number of compatible spheres over the Kleene theory — via the bijection boundaries  $\leftrightarrow$  fiber-minimal elements. We do not know a direct structural link between the two interpretations beyond the count; both “remove two elements”, but different ones ( $\{0, 1\}$  versus  $\{\delta_\top, 1\}$ ).

**Remark 9.4** (Boolean and distributive). For the Boolean theory ( $x \vee \neg x = 1$ ), the representation collapses to functions on the *consistent* points (all pairs  $(0, 1)$  or  $(1, 0)$ ): every point is a face point, no invisible points remain, and all fibers are singletons — boundaries determine their elements, as for the bare bounded-distributive theory (see the proof of Theorem 10.2). With Kalman’s theorem, the table of §1 is therefore a complete classification of boundary-fiber behaviour over the subvariety lattice.

**9.1. The Kleene defect and coherence theory.** We record how §6 specializes. Throughout,  $r(g) = g \setminus A_g$  where  $A_g$  is the set of minimal invisible points of  $g$ ; by Theorem 9.2,  $A_g \subseteq \{\perp, \top\}$ , and indeed  $A_g = \{\top\}$  iff  $g = \delta_\top$ ,  $A_g = \{\perp\}$  iff  $g = 1$ .

**Theorem 9.5.** *Over the free Kleene algebras:*

- (1) (defect formula) *for every  $g$ ,  $r(g)$  is the least element of the boundary fiber of  $g$  in one deletion step;*
- (2) (coherence dichotomy) *for every substitution  $f$  (inducing  $F$  on unmixed points — substitutions preserve unmixedness) and every  $g$ :  $r(f^*g) = r(f^*r(g))$  iff  $F(\text{Face}(U_m)) \cap A_g = \emptyset$ ;*
- (3) (polar exceptional locus) *consequently the incoherent pairs  $(f, g)$  have  $g \in \{\delta_\top, 1\}$ : for each substitution, at most two elements — independent of  $n$  — fail to transport coherently, in contrast with the positive-fraction exceptional loci of the De Morgan case (Remark 6.5).*

*Proof.* (1) and (2) are the proofs of Theorems 6.1 and 6.3 verbatim over the poset  $U_n$ , with Lemma 9.1 supplying the two ingredients used there

(faces see exactly face points; a face point lies between comparable invisible points). Preservation of unmixedness by substitutions is the semantic fact that lower/upper approximations are computed uniformly:  $F(v_{\text{low}}) = (t(v))_{\text{low}}$  for the three-valued evaluation  $t$  of the substitution. (3) is immediate from  $A_g \subseteq \{\perp, \top\}$  and the identification of the two elements with nonempty defect.  $\square$

**Remark 9.6.** Thus the Kleene theory is well-behaved “up to two poles”: the corrected transport is automatically coherent for every source element outside  $\{\delta_{\top}, 1\}$  — the only possible source-side coherence defects occur at the two poles — and even there the deviation is a single toggled pole, corrected inside a two-element fiber by the median term of Theorem 5.1 (whose Kleene instance is the same term). (As in Proposition 6.4, this is a statement about coherence, not about plain naturality: already for  $f^*(x) = x \wedge \neg x = \delta_{\top}$  and  $g = x$  one has  $r(f^*g) = 0 \neq \delta_{\top} = f^*(rg)$ .) It is instructive that the two exceptional elements are precisely the witnesses of the failure of excluded middle surviving the Kleene law:  $\delta_{\top} = \bigwedge_i (x_i \wedge \neg x_i) \neq 0$  and its dual. The Kleene law tames, but cannot eliminate, the no-LEM elements.

## 10. COMPLETENESS: FIBER BEHAVIOURS ACROSS SIGNATURES

The trichotomy of the introduction classifies fiber behaviour over the subvarieties of De Morgan algebras. One may still ask whether *weaker signatures* — interval theories that drop the involution or the connections — realize intermediate behaviours. They do not.

**Lemma 10.1.** *An interval signature (a subset of  $\{\wedge, \vee, \neg\}$  over the bounds  $0, 1$ ) containing  $\neg$  and at least one connection defines the other connection by De Morgan’s laws. Hence the signature classes are:  $\emptyset$ ,  $\{\wedge\}$ ,  $\{\vee\}$ ,  $\{\wedge, \vee\}$  (the bounded-distributive theory),  $\{\neg\}$ , and the full De Morgan signature.*

**Theorem 10.2** (completeness). *Over every interval signature not containing both the involution and a connection, all boundary fibers of the free algebras are singletons. Consequently the realizable fiber behaviours of interval theories are exactly three: trivial (bounded-distributive, Boolean, and all weaker signatures), two polar fibers (Kleene), and Sperner-extremal (De Morgan).*

*Proof.* By Lemma 10.1 four signature classes remain below the full one. *Bounded-distributive:* the free bounded distributive lattice on  $n$  generators is the lattice of monotone functions on  $\{0, 1\}^n$ ; every evaluation point has a first coordinate in  $\{0, 1\}$ , so an element is determined by its two faces  $x_1 := 0, 1$  — fibers are singletons. (The same two-face decomposition handles the *Boolean* algebra of all functions.) *Meet-only* (dually join-only): the free bounded  $\wedge$ -semilattice consists of 0 and the conjunctions  $\bigwedge_{i \in S} x_i$ ; the faces of a conjunction determine  $S$  (whether  $i \in S$  is read off the  $x_i := 0$  face), and distinguish 0 from 1. *Involution-only:* the free algebra is  $\{0, 1, x_1, \neg x_1, \dots\}$ , and the faces of  $x_i, \neg x_i$  (constants 0, 1 in opposite positions) separate all

elements. *Bare*: immediate. In each case no evaluation point is invisible to the faces; the nontrivial behaviours require literals to decouple ( $\neg$ , giving independent pairs) *and* to combine ( $\wedge$  or  $\vee$ ), and once both are available the algebra is a De Morgan algebra, where Kalman’s theorem [12] and the fiber theorems (Theorems 3.4 and 9.2) leave exactly the three grades.  $\square$

**Remark 10.3.** The moral sharpens the through-line of the introduction: *fiber freedom is born only from the interaction of the involution with a connection, and once born it is exhausted by Kalman’s three grades*. In particular no interval *theory* interpolates strictly between the Kleene and De Morgan rows — any graded interpolation must come from non-equational structure (restricted substitution classes, cf. the coherence analysis of §6), not from the algebra.

## 11. THE CENSUS

All finite combinatorial claims were verified by exhaustive enumeration in the monotone-function representations (not by a proof assistant: the verifications are reproducible computations, with the mathematical proofs given in the text); the scripts accompany the artifact (see below) and are independently reproducible from the pinned release.

**11.1. De Morgan.** Through  $n \leq 3$  (at  $n = 3$  the full algebra of  $M(6) = 7,828,354$  elements, with boundaries grouped by their 6-tuples of faces); writing  $B(n)$  for the number of realizable boundaries:

| $n$ | $ \text{DM}_n  = M(2n)$ | $B(n)$    | fiber sizes (size: count)                  | max |
|-----|-------------------------|-----------|--------------------------------------------|-----|
| 1   | 6                       | 4         | 1:2, 2:2                                   | 2   |
| 2   | 168                     | 82        | 1:28, 2:38, 4:16                           | 4   |
| 3   | 7,828,354               | 4,829,136 | 1:2698202, 2:1747364,<br>4:358284, 8:25286 | 8   |
| 4   | $M(8)$                  | —         | (not enumerated; see below)                | 64  |

At  $n \leq 3$  the maximum is  $2^{\lfloor n/2 \rfloor} = 2, 4, 8$ ; the Boolean complement formula of Theorem 3.4 succeeded in all 2,130,934 non-singleton fibers at  $n = 3$ , with no failures. At  $n = 4$  full enumeration is out of reach ( $M(8) \approx 5.6 \times 10^{22}$ ), but the realizability construction of Theorem 3.4 applied to the middle layer of the diagonal cube (an antichain of size  $\binom{4}{2} = 6$ ) produces an explicit fiber isomorphic to  $2^6$ , all of whose 64 members were verified by machine to share their boundary; with the Sperner upper bound this settles  $\max = 64$  at  $n = 4$ . The column  $B(3)$  was computed twice, independently: once by direct fiber-grouping, and once as the antichain count of Theorem 4.3. The identity of Corollary 3.6 checks:  $28 + 2 \cdot 38 + 4 \cdot 16 = 168$  and  $2698202 + 2 \cdot 1747364 + 4 \cdot 358284 + 8 \cdot 25286 = 7828354$ .

The defect formula of Theorem 6.1 was verified on all elements at  $n = 2, 3$  against brute-force fiber minima (zero discrepancies); the coherence dichotomy on all of  $\text{DM}_2$  for five generating substitutions (81 incoherent instances, counted independently as  $23 + 23 + 24 + 11$ ), and on all of  $\text{DM}_3$



for a diagonal, a connection, and the reversal — conditions (1), (2), (3) of Theorem 6.3 carve out identical subsets, of sizes 1,133,756, 1,133,756, and 0 respectively.

None of the sequences

$$B(n) = 4, 82, 4829136, \dots \quad (\text{nor } 2, 28, 2698202 \text{ and } 2, 16, 25286)$$

matched any entry of the OEIS [15] as of August 2026; we intend to contribute them. Interval-size statistics for the ambient monotone-function lattices, without the boundary grouping, were computed by Pawelski [17].

**11.2. Kleene, and the weaker signatures.**  $|\text{Mon}(U_n)| = 6, 84, 43918$  against the Berman–Mukaidono spectra; fiber censuses  $\{1^2, 2^2\}$ ,  $\{1^{80}, 2^2\}$ ,  $\{1^{43914}, 2^2\}$  confirming Theorem 9.2 and the identity  $B_{\text{KL}}(n) = |\text{KL}_n| - 2$ ; the defect formula on all 84 elements of  $\text{KL}_2$  against brute-force fiber minima; and the coherence dichotomy on all of  $\text{KL}_2$  for five generating substitutions — the conditions of Theorem 9.5(2) coincide exactly, with incoherent loci of sizes 2, 2, 0, 2, 0 (both connections, reversal, diagonal,  $x \mapsto x \wedge \neg x$ ), each consisting of the two polar elements. Assertions that substitutions preserve unmixedness and that defects stay within  $\{\perp, \top\}$  were checked at every step. For Theorem 10.2 the singleton-fiber claims were enumerated as well: bounded-distributive  $n \leq 3$ , meet-only  $n \leq 3$ , involution-only  $n = 2$  — all fibers singleton. At  $n = 2$  the boundary counts of the Kleene and De Morgan theories coincide (82): the mixed points of  $Q_2$  are the two non-polar diagonals, so the two face-point posets agree — a low-dimensional accident that disappears at  $n = 3$  (43,916 versus 4,829,136).

**Question 11.1.** Is there a closed form or efficient recursion for  $B(n)$ , i.e. for the number of antichains of the face-point poset of  $Q_n$ ? The poset is a  $2n$ -cube minus its diagonal  $n$ -cube; its antichain count interpolates between Dedekind-type growth and the structure imposed by the excised diagonal.

**Question 11.2.** What is the distribution of  $|D|$  over fibers as  $n \rightarrow \infty$  — equivalently, of the number of free diagonal points of a random boundary? At  $n = 3$  about 65.5 % of the elements lie in non-singleton fibers (and about 34.5 % in singleton ones); among the 4,829,136 fibers themselves, about 44.1 % are non-singleton.

## 12. OUTLOOK

Three directions seem worth recording. (1) With the equational axis closed by Theorem 10.2, graded interpolations between the Kleene and De Morgan rows can only arise from *non-equational* data — restricted classes of substitutions, as in the coherence analysis of §6; a classification of coherence behaviour over natural generator classes seems within reach. (2) The idempotent completions of Theorem 7.2 invite a boundary-fiber theory over finite projective De Morgan and Kleene algebras themselves — whether the polar cells of the Kleene case survive projectivization is open. (3) In the

cubical-sets reading that motivated this work, the results equip the strict layer of De Morgan cubical sets with canonical least fillers, Boolean fiber decompositions, and strict cylinders; their homotopical consequences are studied in a separate line of work (see the artifact below).

### Declarations.

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Data availability. The enumeration scripts and computational certificates supporting the finite enumerations, and the machine-checked counterparts of Theorems 3.4 and 5.1 in a cubical proof kernel, accompany the artifact repository <https://github.com/r-shibusawa/cubical-strict-j> (release v1.12.0, commit f15e4864), archived at the version DOI 10.5281/zenodo.21785412 (concept DOI 10.5281/zenodo.21405961). The relevant scripts are `dmfiber.py`, `dmfiber3.py`, `dmfiber_boundaries.py`, `dmdefect.py` and `kleene.py`.

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Use of AI tools. During the preparation of this work the author used an AI assistant to draft and revise the manuscript text and to develop and verify the computations; the author reviewed and edited all content and takes full responsibility for it.

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